

Verify the result. Keep the readings private.



Measure

Capture the source record.



Prove

Apply the registered policy.



Verify

Share the public evidence.

Private measurements → a result another organization can check

A report asks for trust. Evidence makes it checkable.

Operator

Holds sensitive measurement records.

Reviewer

Needs the condition, date, and result.

BACCHIRI

Binds that result to a registered policy.

Initial use case: cross-organization measurement reporting

Bind the Device and policy first

G01

New GUI recording pending

Capture: Device Workflow → registered Device and applied policy

Shot placeholder. No recorded action or result is shown.

Controlled browser demo · synthetic measurements

Device

Registered identity

Policy

Public bounds and validity

Authority

Controlled by the Device

One day. Twenty-four hourly slots.

G02

New GUI recording pending

Capture: selected date, record count, and private-input availability

Shot placeholder. No recorded action or result is shown.

Controlled browser demo · synthetic measurements

Select

A completed operational day

Keep private

Readings and proof opening

Preserve gaps

Missing hours remain NO DATA

The Device authorizes. The service schedules.

G03

New GUI recording pending

02:00 JST starts scheduled processing; confirmation can come later

Shot placeholder. No recorded action or result is shown.

Authorize

Approve the Device payload

Queue

Track the same Proof Job ID

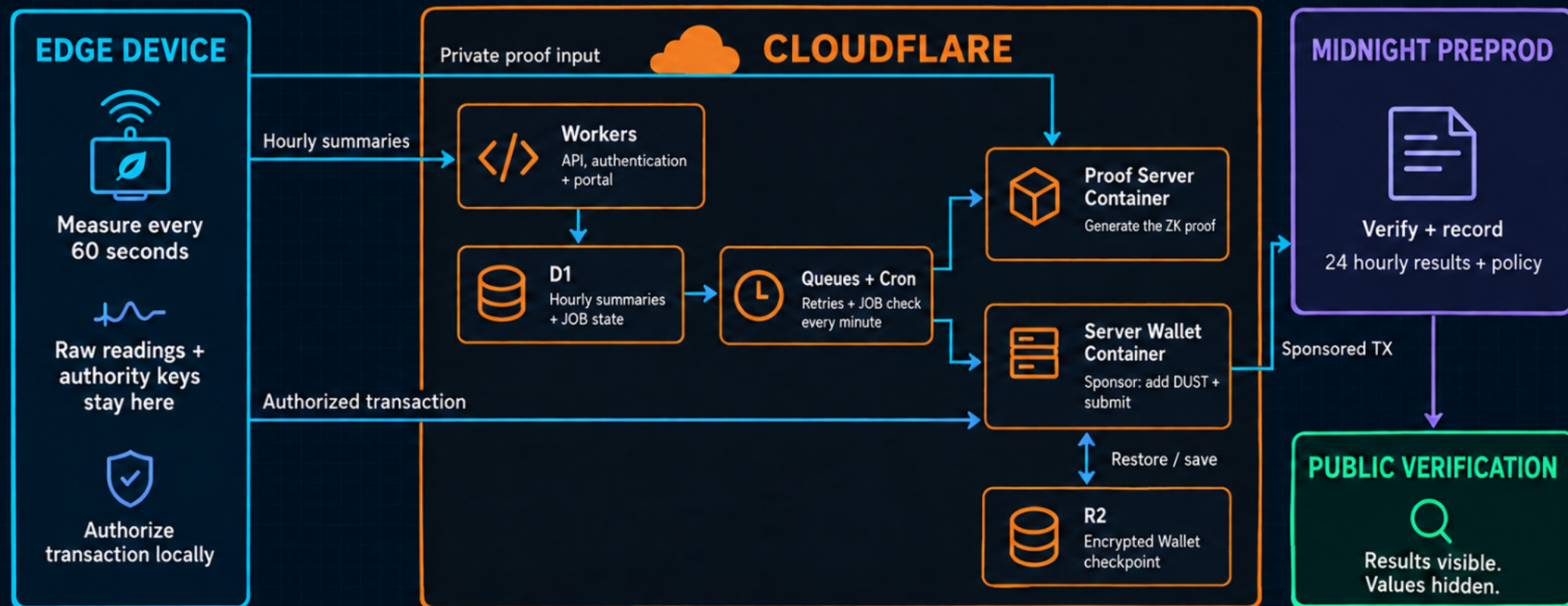
Wait

Pending is not verified

Controlled browser demo · synthetic measurements

CONTINUOUS MEASUREMENT. SCHEDULED PROOF.

What runs on Cloudflare



🕒 **02:00 JST • START PROCESSING THE PREVIOUS DAY**

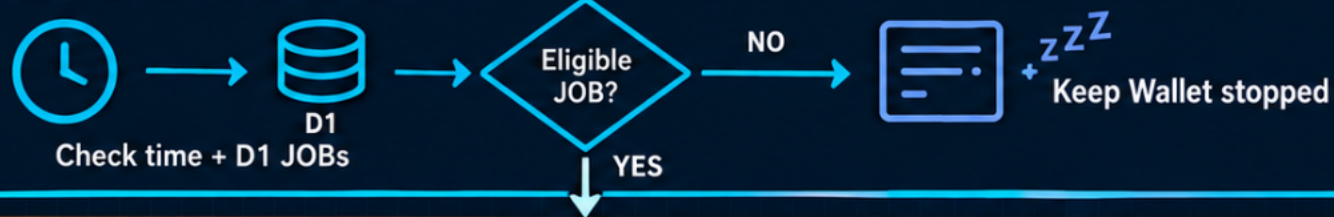
Trusted backend receives private proof input. Sensor values are hidden from public viewers.

Architecture illustration · field device path

THE WALLET RUNS ONLY WHEN WORK IS DUE

Scheduled mode · daily processing starts at 02:00 JST

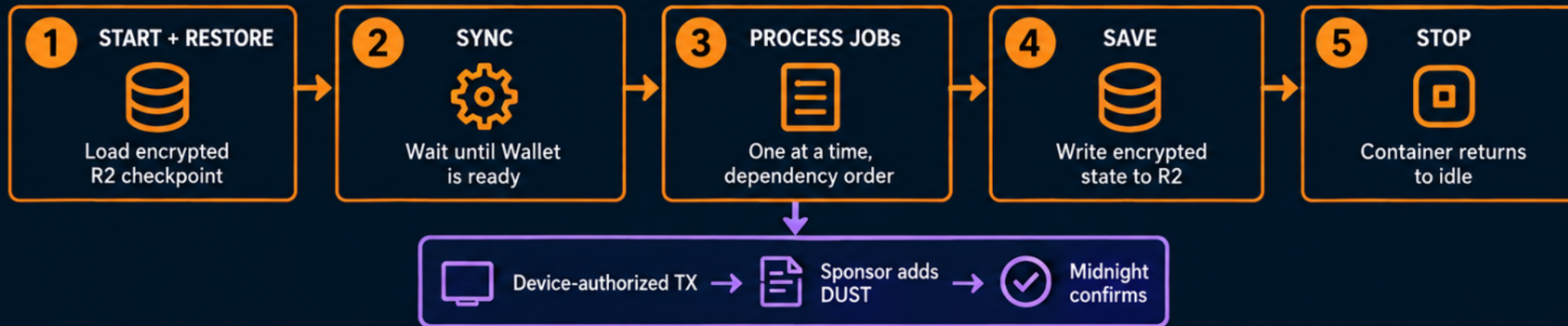
WORKER · EVERY MINUTE



Next eligible batch: check again

New JOBs after the daily cutoff wait for the next run.

SERVER WALLET CONTAINER · ONLY WHILE PROPROCESSING



SENSOR COLLECTION CONTINUES EVERY 60 SECONDS

02:00 is the processing start. Completion follows synchronization, proof generation and confirmation.

Architecture illustration · field device path

Return to the same job after processing

G04

New GUI recording pending

Later capture · elapsed waiting time is omitted

Shot placeholder. No recorded action or result is shown.

Later capture · elapsed waiting time is omitted

Match

Same Device, day, and job

Confirm

Confirmed transaction state

Carry forward

Transaction hash for verification

A reviewer starts with the transaction hash

G05

New GUI recording pending

No Device session required for the public verifier

Shot placeholder. No recorded action or result is shown.

Look up

The same confirmed transaction

Inspect

Day, hourly results, policy

Compare

Public Midnight evidence

Controlled browser demo · synthetic measurements

A precise proof has a clear boundary

Established

The relation between committed input, policy, and result.

Kept private

Sensor values in the public verification view.

Still trusted

Physical accuracy, complete sampling, source aggregation, and the backend.

Cryptographic evidence supports a claim with a defined scope

Turn a working proof flow into a measured partner pilot

Demonstrated

Core proof flow on Midnight Preprod.

Implemented

Field collection and scheduled Cloudflare processing.

Next

Validate reliability, review effort, and willingness to pay.

Seeking a measurement-reporting workflow and a pilot partner